SPRINT 3 - TEAMWORK PLAN

Sprint start date: 19 October 2026
Sprint end date: 1 November 2026
Team level: Clear, practical instructions for every team member

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1. What are we trying to achieve?

Our situation

We have already completed the main preparation work:

- The energy dataset has been cleaned and frozen.
- The rules for preparing data for machine learning have been documented.
- The train, validation and test strategy has been defined in Task 6.
- Our team has built its first forecasting model.

We are now ready to find out whether our model can actually predict energy consumption accurately.

Our goal for Sprint 3

By the end of this sprint, we want to answer four simple questions:

1. **Can our first model predict energy consumption reasonably well?**
2. Is it better than simple prediction methods?
3. Which model gives the best results, and where does it make mistakes?
4. What should we improve during Sprint 4?

Important: Our goal is not to build the most complicated model. Our goal is to find a model that works well and prove it using reliable tests.

2. The rules everyone must follow

Before starting, everyone must understand these five rules.

Rule 1 - Never change the frozen dataset.
Create a separate working copy if you need to prepare data for a model.

Rule 2 - Test models fairly.
Models must use the same evaluation periods and, wherever possible, the same observations.

Rule 3 - Never look at the test results to improve a model.
Use the validation period to choose or improve a model. Use the test period only for the final evaluation.

Rule 4 - Never invent missing energy values.
If an energy value is missing or unreliable, do not replace it with zero. Record the problem.

Rule 5 - Keep the two historical periods separate.
The 2020–2023 data and the 2025 data must be evaluated separately because there is a long gap between them.

Which dates do we use?

For the main evaluation, Task 6 proposes the following split:

Period	Dates	What we use it for
Training	19 August – 21 September 2025	Build the model and calculate training statistics
Validation	22–28 September 2025	Compare models and choose the best one
Test	29 September – 5 October 2025	Measure the final performance

These dates must be checked against the available data before running the experiments.

3. The six tasks

TASK 1 - Build two simple prediction methods

Why are we doing this?

Before deciding that our machine-learning model is good, we need to compare it with simple methods. A complex model is not useful if a simple method gives equally good predictions.

What must we do?

Build two basic forecasting methods.

Method A - Use the same hour last week

For example, to predict Monday at 10:00, use the observed value from the previous Monday at 10:00.

Steps:

1. Take each prediction time.
2. Find the value from exactly 168 hours earlier.
3. Use that value as the prediction.
4. If the earlier value is unavailable, mark the prediction as unavailable.

Method B - Use the usual value for that weekday and hour

For example, to predict Tuesday at 14:00, calculate the median of the available training values for Tuesdays at 14:00.

Steps:

1. Use training data only.
2. Group the values by weekday and hour.
3. Calculate the median for each group.
4. Use the corresponding median as the prediction.

Do not use validation or test values to calculate these medians.

Run the methods for the agreed targets and prediction horizons wherever possible.

Deliverable:

A file containing the baseline predictions and their MAE and RMSE results.

How do we know the task is finished?

We have two simple methods that produce predictions and can be compared with our first model.

Can this be done in parallel?

Yes. One person can build Method A and another can build Method B.

TASK 2 - Test the first machine-learning model

Why are we doing this?

We already have a first model. Now we need to check whether it works on data it has not used to learn.

What must we do?

1. Find the code used to build the first model.
2. Make sure another team member can run it.
3. Write down which target it predicts, which input variables it uses and how the model is configured.
4. Train it using the training period.
5. Measure its performance on the validation period.
6. Check that no future target values are used as inputs.
7. Make sure that any preprocessing learned from data is fitted using training data only.
8. Save the predictions and evaluation results.

Do not use the test period to change the model.

Deliverable:

A reproducible script or notebook, the saved predictions and the validation results.

How do we know the task is finished?

Another team member can run the model and obtain the documented results using the same settings.

TASK 3 - Test one alternative model

Why are we doing this?

Our first model may not be the best option. We should test one reasonable alternative and see whether it improves the predictions.

What must we do?

1. Check which algorithm was used for the first model.
2. Choose one alternative that the team can implement and understand.
3. LightGBM is a possible choice if it is not already the first model and the team has the time and tools to use it.
4. Use the same target, training period, validation period and prediction horizons.
5. Save the predictions and calculate MAE and RMSE.
6. Record any problems that prevent a fair comparison.

Prophet is optional. Do not start several extra models if doing so prevents the team from evaluating the existing models properly.

Deliverable:

Results from one alternative model, or a short explanation of why testing it was not feasible.

How do we know the task is finished?

We have results that can be compared fairly with the two baselines and the first model.

Can this be done in parallel?

Yes. It can run alongside Tasks 1 and 2.

TASK 4 - Compare the models and find their mistakes

Why are we doing this?

We need evidence to decide which model works best. We also need to understand where the predictions are unreliable.

What must we do?

1. Collect the results from Tasks 1, 2 and 3.
2. Put them in one comparison table.

3. Use **MAE** as the main error metric. Lower MAE means smaller average absolute errors.
4. Use **RMSE** as a second metric. It gives more weight to large errors.
5. Compare models on the same valid observations wherever possible.
6. Identify the model with the best validation results.
7. Once the model setup is fixed, evaluate the selected models on the test period.
8. Look at the largest errors. Check whether they occur at particular hours, days, zones or periods with data-quality problems.
9. For 24-hour forecasts, report the overall result and, where feasible, how errors change from hour 1 to hour 24.

Do not use MAPE as the main metric because percentage errors can become misleading when actual values are zero or close to zero.

Deliverable:

One comparison table and a small set of clear graphs or examples showing the biggest prediction errors.

The table should contain:

Target	**Forecast horizon**	**Model**	**MAE**	**RMSE**	**Period**
energy_Wh	+1 hour	Baseline A	To calculate	To calculate	Validation/test
energy_Wh	+1 hour	Baseline B	To calculate	To calculate	Validation/test
energy_Wh	+1 hour	First model	To calculate	To calculate	Validation/test
energy_Wh	+1 hour	Alternative model	To calculate	To calculate	Validation/test

Add rows for other horizons and targets only when reliable results are available.

How do we know the task is finished?

We can answer: Which model performed best? Was it better than the simple baselines? What types of errors remain?

Can this be done in parallel?

Not fully. The comparison depends on the predictions from Tasks 1, 2 and 3. However, one person can prepare the results table while another prepares the graph templates.

TASK 5 - Show how uncertain the predictions are

Why are we doing this?

A prediction is not always exact. People using the forecasts should understand that the actual energy consumption may be higher or lower than the predicted value.

What must we do?

1. Start with the best-supported model selected using validation results.
2. Choose a simple, explainable method to estimate a lower and upper prediction limit.
3. Apply the method without using test results to tune it.
4. Check how often the actual values fall between the lower and upper limits.
5. Report the percentage of actual observations covered by the interval.
6. Explain that a wider interval means greater uncertainty.

Do not claim that the prediction band is reliable unless its performance has been measured.

Deliverable:

A graph showing predictions and lower/upper limits, together with a short coverage summary.

If this task cannot be completed properly during the sprint, deliver a short note describing the proposed method, what remains to be done and why.

How do we know the task is finished?

We can show the prediction interval, explain how it was created and report how often it contained the actual value or clearly document what is still missing.

Can this be done in parallel?

The team can research methods in parallel. The final implementation should wait until there is a sufficiently evaluated candidate model.

TASK 6 - Prepare the final report and Sprint 4 handover

Why are we doing this?

The project must be understandable to someone who did not build the models. We also need to know what to do next.

What must we do?

Prepare a short report in English containing:

1. The objective of the forecasting project.
2. The data and evaluation periods used.
3. The models tested.
4. The comparison table with MAE and RMSE.
5. The best-performing model and whether it beats the baselines.
6. The main prediction errors and data limitations.
7. The prediction-band results, if available.
8. A recommendation for Sprint 4.

Organise the code, configuration, predictions, graphs and results so another team member can reproduce the experiments.

Do not modify the frozen source dataset.

Deliverable:

A final report, reproducible code or notebook, results table, graphs, experiment record and a list of recommended next steps.

How do we know the task is finished?

The team can present the results clearly, another member can reproduce the main evaluation, and the next sprint has a concrete starting point.

Can this be done in parallel?

Yes. One person can maintain the experiment record throughout the sprint, while another prepares the report structure. The final conclusions must wait for Task 6.

4. What must we have at the end of Sprint 3?

The sprint is successful if the team has:

- A confirmed evaluation setup.
- A separate modelling dataset, with data limitations recorded.
- Two working baseline methods.
- A reproducible evaluation of the first model.
- Results from one alternative model, if feasible.
- A comparison table using MAE and RMSE.
- An analysis of the main prediction errors.
- Prediction uncertainty results, or a documented plan to finish them.
- A concise English report and organised experiment files.
- A clear recommendation for Sprint 4.

Minimum priority if time is limited: Complete Tasks 1–2 and 4 first. Task 3 is important if feasible; Task 5 can be carried over if it cannot be completed reliably. Do not sacrifice a fair evaluation just to test more algorithms.

6. What comes next?

Sprint 3 is mainly about proving whether the forecasting model is useful.

Sprint 4 can then build on those findings to investigate unusual energy consumption and possible energy waste. The team should not claim that it can detect waste automatically until that task has been designed and evaluated separately.

Final message to the team: Work in small, clear steps. Save your code and results. Record problems instead of hiding them. A simple model with trustworthy results is more valuable than a complicated model that nobody can explain.